Chromatin remodeling of type-1 interferon-inducible genes is associated with viral innate immune response in a sex-dependent manner.

Newborn males and females exhibit distinct immune responses to viral infections, leading to sex-based differences in susceptibility to viral diseases. This study investigates the role of transcriptional regulation and chromatin remodeling of type I interferon (IFN)-inducible genes as potential mechanisms underlying these sex-associated disparities in early-life innate immune responses. Researchers studied the immune response in newborn cells by comparing the expression of IFN-I genes and the configuration of the chromatin between males and females. Bone marrow dendritic cells (BMDCs) from 7-day-old male and female mice were collected and cultivated with GM/CSF. BMDCs were exposed to imiquimod for 16 hours and analyzed by RT-qPCR, which showed a higher expression of pro-inflammatory genes like IFNB in female BMDCs. The study demonstrated that female mice have greater T-cell IFN-γ production than male mice. Early-life RSV infection led to long-term alterations in BMDCs in a sex-dependent manner. At 4 weeks post-neonatal Respiratory Syncytial Virus (RSV) infection, BMDCs from female mice (EL-RSV/F) showed significantly higher expression of Ifnb and Il12a upon in vitro RSV exposure compared to BMDCs from male mice (EL-RSV/M) and naïve age-matched controls. EL-RSV/F BMDCs also exhibited increased IFNAR1 expression and induced higher IFN-γ production by CD4+ T cells. In vivo experiments demonstrated functional differences between EL-RSV/F and EL-RSV/M BMDCs. Transfer of EL-RSV/F BMDCs into naïve adult mice reduced viral gene expression upon RSV challenge, while transfer of EL-RSV/M BMDCs increased pathology and Th2/Th17 responses. ATAC-seq analysis revealed that female mice had more accessible chromatin than male mice at 4 weeks post-neonatal RSV infection. This increased accessibility allows for more efficient activation of gene transcription in response to infection. Transcription factors such as JUN, STAT1/2, and IRF1/8 were predicted to have binding sites in these accessible regions in female cells after early-life infection.The study also found that human cord blood-derived monocytes displayed a similar sex-associated chromatin landscape, with female-derived monocytes showing more accessibility in type-I immune genes. This research highlights sex-based differences in innate immunity, controlled epigenetically and amplified by early-life infection in females via type-I immunity. These findings enhance our understanding of sex-associated immune differences and could have important implications for developing targeted therapies. The underlying epigenetic mechanisms, including the role of sex hormones and the X chromosome, merit further exploration in future studies.

Differences in viral innate immune reponse in a sex-based manner